

Notes: Coval and Moskowitz (JF, 1999)

Home Bias at Home: Local Equity Preference in Domestic Portfolios

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1 Big picture

- **Question.** Do investors prefer geographically nearby stocks even within one country? More specifically, do U.S. professional money managers tilt domestic equity portfolios toward firms headquartered near the manager?
- **Motivation.** The international “home bias” puzzle says investors hold too much domestic equity relative to the gains from global diversification. The usual explanations mix two different forces:
 - **national-border frictions:** exchange rates, taxes, regulation, language, culture, sovereign risk;
 - **distance frictions:** information, familiarity, travel/contact costs, and possibly hedging demand for locally consumed goods.

The paper isolates the second force by studying portfolio choice *inside* the United States, where national borders, currencies, and many legal frictions are absent.

- **Main finding.** U.S. investment managers exhibit a strong local-equity preference. The average manager’s holdings are about 160–184 kilometers closer than the average stock in the relevant benchmark portfolio. In percentage terms, holdings are about 9–11 percent closer than the benchmark.
- **Interpretation of the title.** The paper documents “home bias at home”: even when all stocks are domestic U.S. stocks, investors still behave as though distance matters.
- **Key empirical object.** The local-bias statistic compares the distance from a manager to her actual portfolio with the distance from the same manager to a benchmark portfolio. The benchmark can be equal-weighted, value-weighted, or style-specific.
- **What makes the result interesting.** The paper does not merely show that managers in Boston or New York hold many nearby stocks. It asks whether portfolio-weight deviations from a benchmark are correlated with distance. This matters because firms and managers are both geographically clustered.
- **Main economic result.** Local bias is strongest for firms that are:
 - small,
 - highly levered,
 - producers of nontraded goods or firms without foreign sales.

These are exactly the types of firms for which local information may be most valuable.

- **Main interpretation.** The results support an information-based view of local bias. Local investors may have easier access to firm information through local media, employees, suppliers, customers, executives, and informal networks.
- **Relation to international home bias.** By extrapolating the estimated domestic distance effect to global distances, the authors argue that distance alone could account for roughly one-third of the observed U.S. home-country equity bias.
- **Asset-pricing connection.** Since small and highly levered firms are also central to size and distress-related asset-pricing patterns, local ownership may be linked to cross-sectional return

anomalies. The paper does not fully solve those anomalies, but it suggests that information asymmetry and geography are part of the mechanism.

- **Caveat.** The paper is primarily about *portfolio holdings*, not realized performance. It documents who holds what and how holdings relate to distance and firm characteristics. Performance of local holdings is mostly left for related/follow-up work.

2 Data and measurement (key objects)

2.1 Main data sources

- **Investment manager holdings.** The main source is Nelson’s 1996 *Directory of Investment Managers*. It provides 1995 holdings data for large U.S. money managers and reports each manager’s city and state.
- **Firm headquarters.** Firm headquarters locations are obtained from Compact Disclosure.
- **Coordinates.** The authors match both manager locations and corporate headquarters to latitude and longitude using the U.S. Census Bureau’s Gazetteer Place and Zip code Database.
- **Firm characteristics.** Accounting and market variables come from COMPUSTAT and Compact Disclosure.
- **Unit of observation for portfolio tests.** A fund manager/fund portfolio with top holdings in U.S.-headquartered companies.
- **Unit of observation for regression tests.** A manager-holding pair: manager i holding stock j .

Interpretation. The data allow the authors to measure physical distance between the portfolio manager and every stock she holds or could have held. The paper turns geography into a measurable portfolio-choice variable.

2.2 Sample construction

- The sample is restricted to U.S.-based investment managers investing primarily in U.S. equities.
- A fund is classified as primarily U.S. equity if at least five of its top 10 holdings are U.S.-headquartered firms.
- Index funds are removed because the paper wants managers that can actively choose holdings.
- Funds and firms in Alaska, Hawaii, and Puerto Rico are excluded so that remote locations do not mechanically dominate distance calculations.
- Investments by one manager in another manager’s fund are ignored.
- The investable universe is defined as companies held by at least one mutual fund and covered by the required data sources.

Final sample.

- 1,189 investment managers.

- 2,183 different U.S. equity funds.
- 2,736 different U.S. companies.
- About \$1.789 trillion invested in U.S. equities.
- 18,187 manager-holding observations in the main holding-level regressions.

Important limitation. Nelson reports only the top 10 positions of each investment manager. The paper notes that these positions typically represent about 30 percent of a manager’s total asset value. Thus, the paper measures local bias in the most important holdings, not in every position.

2.3 Manager characteristics

The paper also uses manager-level variables:

- total assets under management;
- whether the manager has branch offices;
- whether the manager is based in New York City;
- reported percentage of research done in-house, on the street, or by consultants/other sources;
- number of companies followed regularly.

Summary facts from Table I.

- 426 managers have branch offices.
- 347 managers are based in New York City.
- Average fund size is about \$820 million; median fund size is about \$149 million.
- Managers report that, on average, 66 percent of research is in-house, 28 percent is street research, and 6 percent is consultant/other research.
- The average manager follows 748 companies regularly; the median is 250.

2.4 Firm characteristics

The firm-level variables used to explain local bias include:

- log market value, $\ln(MV)$;
- leverage, defined as total liabilities divided by total assets;
- current ratio, defined as current assets divided by current liabilities;
- return on assets;
- market-to-book ratio;
- number of employees;
- a tradable-goods dummy equal to one if the firm has positive foreign sales.

Interpretation. These variables distinguish between competing explanations for local bias. If local bias is informational, it should be stronger in firms where outside information is harder to obtain and where local information is more valuable: small firms, distressed firms, and firms selling primarily in local/nontraded markets.

3 Empirical design and specifications (all equations + interpretation)

3.1 Distance between manager and firm

Let manager i have latitude ϕ_i and longitude λ_i , and let firm j have latitude ϕ_j and longitude λ_j . A convenient equivalent version of the paper’s arclength formula is

$$d_{ij} = r \cdot \arccos(\sin \phi_i \sin \phi_j + \cos \phi_i \cos \phi_j \cos(\lambda_i - \lambda_j)), \quad (1)$$

where r is the radius of the earth and d_{ij} is measured in kilometers.

Interpretation.

- d_{ij} is the geographic distance between manager i and firm j .
- A nearby firm has low d_{ij} .
- A distant firm has high d_{ij} .
- Headquarters location is used as the firm’s location. This is imperfect because a firm’s economic activity may be spread across plants, branches, and markets.

3.2 Actual and benchmark portfolio weights

Let:

- h_{ij} be the actual weight that manager i places on stock j ;
- m_{ij} be the benchmark weight of stock j for manager i .

The benchmark can be:

- equal-weighted across all eligible stocks;
- value-weighted by market capitalization;
- class-specific, such as a regional-fund, sector-fund, or small-cap-fund benchmark.

Interpretation. The key idea is not that managers hold the market portfolio. They clearly do not. The null hypothesis is weaker: deviations from the benchmark should not be systematically related to distance.

3.3 Benchmark-weighted average distance

For manager i , the average distance to the benchmark portfolio is

$$d_i^M = \sum_{j=1}^n m_{ij} d_{ij}. \quad (2)$$

Interpretation.

- d_i^M answers: how far away would manager i be from her holdings if she held the benchmark portfolio?
- It normalizes for the manager's location. A Seattle manager is mechanically farther from many listed firms than a Chicago manager.
- Scaling by d_i^M makes local-bias measures comparable across managers.

3.4 Individual-manager local bias statistic

The local bias of manager i is

$$\text{LB}_i = \sum_{j=1}^n (m_{ij} - h_{ij}) \frac{d_{ij}}{d_i^M}. \quad (3)$$

Sign interpretation.

- $\text{LB}_i > 0$: manager i overweights nearby stocks and/or underweights distant stocks relative to the benchmark.
- $\text{LB}_i < 0$: manager i overweights distant stocks and/or underweights nearby stocks.
- $\text{LB}_i = 0$: deviations from the benchmark are unrelated to distance.

Why the sign is positive for local bias. Suppose a manager overweights a nearby stock. Then $h_{ij} > m_{ij}$, so $m_{ij} - h_{ij} < 0$, but d_{ij}/d_i^M is small. Suppose the same manager underweights a distant stock. Then $m_{ij} - h_{ij} > 0$, and d_{ij}/d_i^M is large. The large positive distant-stock terms dominate the small negative nearby-stock terms.

3.5 Aggregate local bias statistic

Stack managers and firms into matrices:

- H : actual portfolio weights, with entries h_{ij} ;
- M : benchmark portfolio weights, with entries m_{ij} ;
- D : distances, with entries d_{ij} ;
- D^M : diagonal matrix whose i th diagonal element is d_i^M ;
- w : manager-weighting vector, with nonnegative entries summing to one.

The aggregate local-bias statistic is

$$\text{LB} = w' \text{diag} \left((M - H) \left[D'(D^M)^{-1} \right] \right). \quad (4)$$

The paper uses two manager-weighting schemes:

- equal weighting of managers;
- value weighting by total assets under management.

Interpretation. The statistic asks whether, on average, managers place more portfolio weight than the benchmark on closer firms. A positive and statistically significant LB means domestic equity portfolios are locally biased.

3.6 Holding-level regression specification

To study which kinds of firms attract local investment, the paper defines a manager-holding local-bias contribution:

$$y_{ij} = (m_{ij} - h_{ij}) \frac{d_{ij}}{d_i^M}. \quad (5)$$

The baseline regression is

$$\begin{aligned} y_{ij} = & \alpha + \beta_1 \ln(\text{MV}_j) + \beta_2 \text{Leverage}_j + \beta_3 \text{CurrentRatio}_j + \beta_4 \text{ROA}_j \\ & + \beta_5 \text{MarketBook}_j + \beta_6 \text{Employees}_j + \beta_7 \text{Tradable}_j + u_{ij}. \end{aligned} \quad (6)$$

Interpretation of coefficients.

- A negative coefficient on $\ln(\text{MV})$ means local bias is stronger for smaller firms.
- A positive coefficient on leverage means local bias is stronger for highly levered firms.
- A negative coefficient on the tradable dummy means local bias is stronger for firms without foreign sales, i.e., firms whose output is more local or less internationally traded.
- The regression is run across manager-holding observations, not simply across firms.

3.7 Firm and manager characteristic regression

The extended specification adds manager characteristics:

$$\begin{aligned} y_{ij} = & \alpha + X_j' \beta + \gamma_1 \ln(\text{ManagerAssets}_i) + \gamma_2 \text{BranchOffice}_i \\ & + \gamma_3 \text{InHouseResearch}_i + \gamma_4 \ln(\text{CompaniesFollowed}_i) + u_{ij}, \end{aligned} \quad (7)$$

where X_j contains the firm characteristics from equation (6).

Interpretation. This specification asks whether local bias is explained by manager type. For example, are small managers more local? Are managers with more in-house research more local? Do managers following fewer companies specialize locally?

3.8 International-distance extrapolation

To connect domestic local bias to international home bias, the authors extrapolate the domestic estimate to global distances. Using the conservative equal-weighted/equal-weighted local bias estimate of 9.32 percent over 1,815 kilometers, they construct distance-adjusted country weights:

$$s_d = s \cdot (1 - 0.0932)^{d/1815}, \quad (8)$$

where:

- s is the country's world-market portfolio share;
- d is the distance from the U.S. investor, using New York City as the reference point;
- s_d is the distance-adjusted weight.

Interpretation. This is not a structural international portfolio model. It is a back-of-the-envelope calculation asking how much of international home bias could be generated by a distance preference similar to the domestic one.

4 Identification and credibility (what makes the estimates convincing)

4.1 Why the domestic setting matters

- International home bias is hard to interpret because foreign investment involves distance and national borders at the same time.
- By staying inside the United States, the authors remove many border-based explanations:
 - exchange-rate risk;
 - cross-country tax differences;
 - sovereign risk;
 - major legal/regulatory differences;
 - language and broad cultural barriers.
- If local bias exists even inside the United States, distance itself must be an important portfolio-choice friction.

Interpretation. The empirical setting is powerful because it turns the home-bias puzzle into a within-country geography question.

4.2 Benchmark logic

The paper's central null is not that every manager should hold the CAPM market portfolio. Instead, the null is:

portfolio-weight deviations from the benchmark should be unrelated to geographic distance.

This is important because active managers deviate from benchmarks for many reasons. The local-bias statistic only treats the deviation as evidence of local preference if the deviation is systematically related to distance.

4.3 Controls for clustering and fund type

- Firms and fund managers both cluster geographically. New York and Boston are especially important financial centers.
- The paper therefore repeats the main tests excluding New York City-based managers.
- The paper also separately examines regional funds, sector funds, small-cap funds, and all other funds.
- For each fund type, the benchmark can be defined using that fund class's investment universe rather than the entire market.

Interpretation. These exercises reduce the risk that the result is caused by a few special fund types or by the fact that many managers and firms happen to be near New York.

4.4 Why firm-characteristic patterns support information

The main local-bias result could reflect familiarity, local loyalty, or marketing. The firm-characteristic regressions sharpen the interpretation.

- Small firms are less widely followed and more opaque.
- Highly levered firms have more uncertain future payoffs, so information about them is more valuable.
- Firms without foreign sales are more likely to sell local/nontraded goods and may be harder for distant investors to evaluate.
- Local bias is strongest exactly in these firms.

Interpretation. The strongest evidence is not merely that managers buy local firms. It is that they buy the types of local firms where a local information advantage should matter most.

4.5 Main limitations

- **Only top holdings.** The data contain only each manager's top 10 holdings, not the full portfolio.
- **Cross-sectional snapshot.** The main data are from 1995. The paper is not a long panel of holdings and performance.
- **Headquarters location.** Headquarters may not equal the firm's true economic location. A firm may have plants, customers, employees, and operations across many states.
- **Local information is inferred.** The paper does not directly observe private information. It infers an information motive from the types of firms that attract local holdings.

- **Professional managers only.** The sample is large investment managers, not households. Individual investors may be even more local, but this paper’s direct evidence is institutional.
- **Portfolio choice, not welfare.** The paper does not fully quantify whether local bias improves performance or reduces diversification.

5 Tables and figure

5.1 Figure 1: Geographic distribution of funds and firms

Read: The figure plots the locations of 1,189 investment managers and 2,736 company headquarters using latitude and longitude.

- Firms and managers are spread across the continental United States.
- Managers are more clustered than firms, especially around New York and Boston.
- The distribution of firms and managers roughly resembles the distribution of U.S. population and economic activity.
- Managers exist in nearly all lower-48 states, except Wyoming and the Dakotas.

Why it matters. The figure motivates the need for a formal statistic. Visual clustering alone cannot tell whether managers prefer local stocks because both managers and firms cluster in economically dense areas.

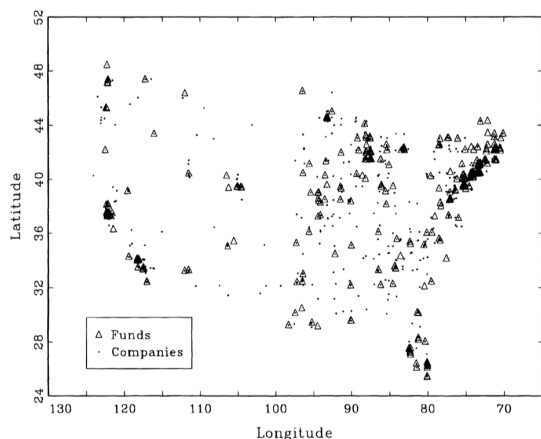


Figure 1. Geographic distribution of U.S. firms and investment managers. Plot of the location of the 1,189 investment fund managers in our sample and the headquarters location of the 2,736 different companies they hold. The horizontal axis contains the actual longitude, converted to degree values, of the fund manager and corporate headquarters location. The vertical axis contains the actual latitude degree values. Latitude and longitude coordinates were obtained from the *U.S. Census Bureau's Gazetteer Place and Zip code Database*. A small amount of random noise was added to each location, so that the mass of funds and companies locating in a given area can be gauged, rather than a single point appearing for New York, for example.

Table I
Summary Statistics of U.S. Investment Managers

All data are from *Nelson's 1996 Directory of Investment Managers*. Summary statistics are reported on funds managed by U.S.-based investment managers that invest primarily in U.S. equities, defined as those funds for which at least five of the top 10 holdings are U.S.-headquartered firms. Fund managers located in Alaska, Hawaii, and Puerto Rico are excluded, and index funds are also removed from the sample. The average percentage of research and number of companies followed regularly are obtained via a survey questionnaire Nelson's sends to each investment manager. Managers are asked to allocate the percentage of research conducted among three categories: (1) in-house, (2) on the street, and (3) consultant/other, as well as report the number of firms they follow on a "regular basis."

Total number of managers:	1189
Managers with branch offices:	426
Managers based in NYC:	347
Number of funds under management:	2183
Total number of different equities held:	2736
Fund size (000's):	
Mean	\$820,000
Median	\$149,000
Min	\$100
Max	\$28,702,000
Total	\$1,789,509,000
Average percentage of research:	
In-house	66%
Street	28%
Consultant/other	6%
Number of companies followed regularly:	
Mean	748
Median	250
Min	8
Max	10000

5.2 Table I: Summary statistics of U.S. investment managers

Read: Table I describes the manager sample.

Statistic	Value
Total managers	1,189
Managers with branch offices	426
Managers based in NYC	347
Funds under management	2,183
Different equities held	2,736
Total fund assets	\$1.789 trillion
Average in-house research share	66%
Average street research share	28%
Average companies followed	748
Median companies followed	250

Interpretation. This is a large institutional sample. The local-bias result is not based on a few retail investors or a small number of funds.

5.3 Table II: Test for local equity preference among all nonindex funds

Read: Table II is the headline result. It compares the average distance from managers to their actual holdings with the average distance from managers to benchmark portfolios.

Weights: Funds-Firms	Holdings	Benchmark	Difference	LB	<i>t</i>
Equal-Equal	1654.18	1814.59	160.41	9.32%	14.28
Equal-Value	1654.18	1830.32	176.15	10.31%	15.93
Value-Equal	1663.09	1833.30	170.21	10.27%	15.21
Value-Value	1663.09	1847.44	184.35	11.20%	16.82
Equal-Equal, ex-NYC	1685.73	1841.03	155.30	8.95%	13.36
Value-Value, ex-NYC	1734.71	1892.32	157.61	9.61%	13.95

Key numbers.

- Actual holdings are roughly 1,654–1,663 km away from the manager.
- Benchmark portfolios are roughly 1,815–1,847 km away.
- Managers invest about 160–184 km closer than the benchmark.
- The local-bias estimate is about 9.3–11.2 percent.
- The *t*-statistics are very large: 14.28–16.82.

Interpretation. U.S. money managers do not hold geographically neutral domestic portfolios. They hold nearby firms more than benchmark portfolios predict.

Why ex-NYC matters. Removing New York City managers still leaves local bias of about 9–10 percent. The result is not just a New York clustering artifact.

5.4 Table III: Local equity preference across fund types

Read: Table III asks whether local bias is driven by special funds such as regional, sector, or small-cap funds.

Table II

Test for Local Equity Preference among All Nonindex Funds

Tests for local bias are reported for the 2,183 nonindex funds in our sample (1,836 excluding funds based in New York City). All combinations of equal-weighted and value-weighted funds and firms are reported, where value weights for firms ($\mathbf{M}$) are the firm's fraction of total market capitalization, and value weights for funds ($\mathbf{w}$) are the fund's fraction of total aggregate asset value under management. Also reported are the components that comprise the local bias statistic, LB . Column 2, for instance, reports the average distance fund managers are from the securities they hold in their portfolios (i.e., $\mathbf{w}'\text{diag}(\mathbf{H}\mathbf{D}')$), where $\mathbf{w}$ is the weighting vector applied to the F funds, $\mathbf{H}$ is the $(F \times n)$ matrix of *actual* portfolio weights each of the F fund managers applies to the n stocks in the economy, and $\mathbf{D}$ is the $(F \times n)$ matrix of distances between fund managers and the headquarters of each stock in the economy. Column 3 reports the average distance fund managers are from their benchmark portfolio (in this case either the equal- or value-weighted index), which is computed as $\mathbf{w}'\text{diag}(\mathbf{M}\mathbf{D}')$, where $\mathbf{M}$ is the $(F \times n)$ matrix of benchmark portfolio weights fund managers are compared to. Column 4 reports the difference between columns 2 and 3, which represents how much closer (in km) managers are actually investing their money relative to their benchmark portfolio. Column 5 reports the LB measure (reported as a percentage), which is $\mathbf{w}'\text{diag}((\mathbf{M} - \mathbf{H})(\mathbf{D}'\mathbf{D}^M)^{-1})$. t -statistics for LB are reported in the last column. Tests are also run excluding funds located in New York city.

Weights: Funds ($\mathbf{w}$)-Firms ($\mathbf{M}$)	Avg. Distance from		Difference	Percentage Bias (LB)	t -stat
	Holdings	Benchmark			
Equal-Equal	1654.18	1814.59	160.41	9.32	14.28
Equal-Value	1654.18	1830.32	176.15	10.31	15.93
Value-Equal	1663.09	1833.30	170.21	10.27	15.21
Value-Value	1663.09	1847.44	184.35	11.20	16.82
Equal-Equal (ex-NYC)	1685.73	1841.03	155.30	8.95	13.36
Value-Value (ex-NYC)	1734.71	1892.32	157.61	9.61	13.95

Table III

Test for Local Equity Preference across Fund Types

Tests are reported for local bias among four subsets of funds: regional, sector, small-cap, and all others. For each subset of funds, the number of managers, average weighted distance from securities held, average distance from the relevant benchmark portfolio, and the difference between these two measures (both in actual km and in percentage terms) are reported. The benchmark portfolio consists of only those stocks being held by at least one fund in the subclass of funds being analyzed. Both equal- and value-weighting schemes are employed to funds and firms, and t -statistics on the local bias measure are provided in the last column.

Fund Type	Weights: Firms-Funds	Avg. Distance from		Difference	Percentage Bias	t -stat
		Holdings	Benchmark			
Regional: ($n = 14$)	Equal-Equal	705.73	1593.25	887.52	53.06	6.55
	Value-Value	983.60	1701.82	718.21	41.79	4.89
Sector: ($n = 85$)	Equal-Equal	1737.04	1801.11	64.07	3.18	0.78
	Value-Value	1672.43	1892.85	220.42	11.62	2.79
Small-cap: ($n = 435$)	Equal-Equal	1755.36	1879.40	124.03	5.69	3.88
	Value-Value	1814.25	1758.25	-55.99	-5.52	-3.69
All others: ($n = 1676$)	Equal-Equal	1625.83	1793.85	168.02	10.24	14.03
	Value-Value	1645.27	1853.81	208.54	12.74	16.80

Fund type and weights	Holdings	Benchmark	LB	t
Regional, Equal-Equal	705.73	1593.25	53.06%	6.55
Regional, Value-Value	983.60	1701.82	41.79%	4.89
Sector, Equal-Equal	1737.04	1801.11	3.18%	0.78
Sector, Value-Value	1672.43	1892.85	11.62%	2.79
Small-cap, Equal-Equal	1755.36	1879.40	5.69%	3.88
Small-cap, Value-Value	1814.25	1758.25	-5.52%	-3.69
All others, Equal-Equal	1625.83	1793.85	10.24%	14.03
All others, Value-Value	1645.27	1853.81	12.74%	16.80

Interpretation.

- Regional funds are strongly local, with local bias around 42–53 percent.
- Sector funds and small-cap funds are less clear because their investment universe is constrained.
- The most important row is “all others”: even after removing regional, sector, and small-cap funds, local bias remains around 10–13 percent.

Why it matters. The local-bias result is broad. It is not just because explicitly regional funds invest locally.

5.5 Table IV: Distance effect on U.S. international portfolio weights

Read: Table IV extrapolates the domestic distance effect to international equity portfolios.

Country	Market	Actual U.S.	Dist.-adj.	Distance
United States	0.478	0.938	0.655	0
Japan	0.265	0.031	0.147	10,918
United Kingdom	0.138	0.011	0.102	5,602
France	0.043	0.005	0.031	5,871
Germany	0.038	0.005	0.028	6,042
Canada	0.038	0.010	0.037	551

Interpretation.

- The market portfolio weight of the United States is 47.8 percent, but U.S. investors actually hold 93.8 percent in U.S. equities.
- The distance-adjusted portfolio raises the U.S. weight to 65.5 percent.
- This does not fully explain home bias, but it moves the portfolio about one-third of the way from the global market portfolio toward observed U.S. holdings.
- The Canada result is intuitive: Canada has a small world-market weight, but because Toronto is close to New York, distance does not reduce its distance-adjusted weight much.

Why it matters. Some portion of international home bias may be ordinary geographic-proximity bias rather than a puzzle that requires country-border frictions.

Table IV
The Distance Effect on U.S. Equity Portfolio Weights

Market capitalization weights of the United States, Japan, United Kingdom, France, Germany, and Canada in the world market portfolio are compared to actual weights assigned by U.S. investors to these countries based on the results from French and Poterba (1991) using 1989 capital flows data, and to distance-adjusted weights calculated by shifting 9.32 percent of a country's remaining market capitalization weight to the United States for every 1,815 km that separate the country from the United States (New York City). Denoting s as each country's share of the world market and d as its distance from the United States, each country's distance-adjusted portfolio share is computed as $s^d = s * (1 - 0.0932)^{d/1815}$

	Portfolio Weights			Distance from NYC (km)
	Market Weight	Actual Weight	Distance-Adjusted	
U.S. (New York)	0.478	0.938	0.655	0
Japan (Tokyo)	0.265	0.031	0.147	10918
U.K. (London)	0.138	0.011	0.102	5602
France (Paris)	0.043	0.005	0.031	5871
Germany (Frankfurt)	0.038	0.005	0.028	6042
Canada (Toronto)	0.038	0.010	0.037	551

Table V
Multivariate Regression (Firm Characteristics)

The dependent variable in the following regressions is the local bias exhibited by a fund manager in a particular holding. The local bias of each holding is calculated as a percentage by multiplying the distance between the manager and each of her holdings by the difference between her benchmark weight applied to each stock and the actual weight she assigned to each stock, divided by the weighted average distance the fund manager is from her benchmark. More formally, $y_{i,j} = (m_{i,j} - h_{i,j})(d_{i,j}/d^M)$, $\forall i,j$, where $m_{i,j}$ is the portfolio weight of stock j in fund manager i 's benchmark portfolio, $h_{i,j}$ is the actual weight fund manager i assigns to stock j , $d_{i,j}$ is the distance between manager i and stock j , and d^M is the weighted average distance between manager i and her benchmark (i.e., $d^M = \sum_j m_{i,j} d_{i,j}$). The regression is run across all fund managers and all of their holdings (18,187 observations) on various firm characteristics. The benchmark portfolio employed is the equal-weighted index of all stocks held by at least one fund. Regressions are run using a Feasible Generalized Least Squares (FGLS) procedure described in Section III.A, where the correlation estimate, ρ (%), from that procedure is reported at the bottom of the table. Coefficient estimates on the firm characteristics are reported, along with their t -statistics in parentheses.

Regression	A	B	C	D
Constant	22.86** (3.50)	16.74* (2.60)	17.48** (2.71)	17.12** (2.33)
ln(MV)	-0.60* (-2.07)	-1.46** (-5.04)	-1.53** (-5.20)	-1.39** (-4.10)
Leverage		42.65** (18.24)	43.16** (18.33)	40.60** (16.93)
Current ratio		0.24 (0.50)	0.26 (0.55)	0.29 (0.62)
Return on assets			-1.25** (-2.91)	-1.21** (-2.84)
Market-book ratio			0.15 (1.23)	0.23 (1.85)
Employees (thousands)				0.02** (4.33)
Tradable dummy				-7.91** (-6.76)
ρ	9.50	8.81	8.79	8.85

*, ** Significant at the 5 and 1 percent levels, respectively.

5.6 Table V: Firm characteristics and local bias

Read: Table V regresses the holding-level local-bias contribution on firm characteristics.

Variable	A	B	C	D
ln(MV)	-0.60	-1.46	-1.53	-1.39
Leverage	—	42.65	43.16	40.60
Current ratio	—	0.24	0.26	0.29
Return on assets	—	—	-1.25	-1.21
Market-to-book	—	—	0.15	0.23
Employees	—	—	—	0.02
Tradable dummy	—	—	—	-7.91

Key results.

Table VI

Multivariate Regression (Firm and Manager Characteristics)

The dependent variable in the following regressions is the local bias exhibited by a fund manager in a particular holding. The local bias of each holding is calculated as a percentage by multiplying the distance between the manager and each of her holdings by the difference between her benchmark weight applied to each stock and the actual weight she assigned to each stock, divided by the weighted average distance the fund manager is from her benchmark. More formally, $y_{i,j} = (m_{i,j} - h_{i,j})(d_{i,j}/d^M) \forall i,j$ where $m_{i,j}$ is the portfolio weight of stock j in fund manager i 's benchmark portfolio, $h_{i,j}$ is the actual weight fund manager i assigns to stock j , $d_{i,j}$ is the distance between manager i and stock j , and d^M is the weighted average distance between manager i and her benchmark (i.e., $d^M = \sum_j m_{i,j} d_{i,j}$). The regression is run across all fund managers and all of their holdings (18,187 observations) on various firm and manager characteristics. The benchmark portfolio employed is the equal-weighted index of all stocks held by at least one fund. Regressions are run using a Feasible Generalized Least Squares (FGLS) procedure described in Section III.A, where the correlation estimate, ρ (%), from that procedure is reported at the bottom of the table. Finally, regressions are run on the full sample of funds, funds not classified as regional (R) or sector funds (S), only small capitalization funds (SC), and all funds not classified as regional, sector, or small-cap. Coefficient estimates on the firm and manager characteristics are reported, along with their t -statistics in parentheses.

Regression:	Full Sample	Non-R,S	Small-Cap	Non-R,S,SC
Constant	27.62** (2.78)	24.23** (2.41)	74.92** (3.24)	33.01** (2.80)
ln(MV)	-1.58** (-4.31)	-1.47** (-3.95)	-2.87** (-2.91)	-2.14** (-4.89)
Leverage	38.53** (15.14)	38.61** (14.93)	17.49** (3.67)	49.16** (15.41)
Current ratio	0.11 (0.22)	0.09 (0.17)	1.79 (0.87)	-0.02 (-0.03)
Return on assets	-1.16** (-2.73)	-1.13** (-2.65)	-0.81 (-1.83)	-8.53** (-4.13)
Market-to-book ratio	0.28* (2.17)	0.32* (2.39)	0.28 (1.46)	0.48** (2.67)
Employees (thousands)	0.02** (3.58)	0.02** (3.70)	0.08* (2.39)	0.02** (2.54)
Tradable dummy	-7.67** (-6.14)	-7.68** (-6.05)	-7.62** (-2.43)	-7.21** (-5.19)
ln(Manager assets)	-0.06 (-0.16)	-0.04 (-0.11)	-1.42 (-1.66)	0.09 (0.21)
Branch office dummy	1.37 (0.93)	1.63 (1.10)	3.64 (1.16)	1.47 (0.87)
% Research in-house	0.01 (0.28)	0.01 (0.40)	0.01 (0.19)	0.01 (0.35)
ln(companies followed)	-0.79 (-1.54)	-0.83 (-1.63)	0.66 (0.55)	-1.04 (-1.84)
ρ	8.60	8.16	6.77	8.58

*, ** Significant at the 5 and 1 percent levels, respectively.

- ln(MV) is negative: local bias is stronger in smaller firms.
- Leverage is large, positive, and highly significant: local bias is stronger in highly levered firms.
- The current ratio is not significant.
- Return on assets is statistically significant but economically small.
- Market-to-book is weak relative to leverage.
- Employees is positive but economically small.
- The tradable dummy is negative and significant: firms with foreign sales are less locally held; firms without foreign sales attract stronger local bias.

Economic magnitudes.

- A one-standard-deviation decrease in log size increases local bias by about 1.5 percent.
- A one-standard-deviation increase in leverage increases local bias by about 10 percent.
- A nontraded-goods firm, proxied by no foreign sales, has about 7.9 percent more local bias than a tradable-goods firm.

Interpretation. The local firms managers overweight are not random local firms. They are firms where information asymmetry is likely to be severe and local information is likely to be useful.

5.7 Table VI: Firm and manager characteristics

Read: Table VI adds manager characteristics and estimates the regression for different samples of funds.

- The firm-characteristic results remain robust: size is negative, leverage is positive, and the tradable dummy is negative across samples.
- Manager assets are not statistically important.
- Branch office status is not statistically important.
- Percentage of in-house research is not statistically important.
- The number of companies followed is economically suggestive but only marginally statistically important. Managers following fewer companies appear more locally biased.

Interpretation. Local bias is not concentrated in one obvious manager type. It appears across a broad range of professional managers, but it is systematically related to the kinds of firms being held.

6 Mechanism and interpretation

6.1 Information advantage story

The authors' preferred interpretation is that local investors may have a comparative information advantage in nearby firms.

Possible channels include:

- local newspapers and business media;
- conversations with employees and managers;
- relationships with suppliers, customers, and local executives;
- local knowledge about demand, reputation, product markets, or financial distress;
- lower travel and monitoring costs.

Why this fits the evidence. The information story predicts stronger local bias where public information is weaker and private information is more valuable. This fits the small-firm, high-leverage, nontraded-goods results.

6.2 Familiarity and behavioral explanations

A softer explanation is that managers are simply more familiar with local firms and therefore more comfortable holding them.

How this differs from information.

- Familiarity does not require true superior information.

- It could reflect psychological comfort, name recognition, local loyalty, or a desire to support the local economy.
- The paper cannot fully rule this out.

Why the paper leans toward information. Pure familiarity would predict local bias in familiar firms generally. The evidence is more specific: local bias is concentrated in firms where information is likely to be most valuable.

6.3 Nontraded-goods hedging story

Another possible mechanism is hedging consumption of local nontraded goods. If investors consume local services or goods whose prices are related to local firms' performance, they may hold local firms as a hedge.

How the paper tests this.

- It uses a tradable-goods dummy based on positive foreign sales.
- Firms with foreign sales are interpreted as more tradable.
- Firms without foreign sales are interpreted as more local/nontraded.

Result. Managers are more locally biased toward firms without foreign sales.

Interpretation. This result is consistent with nontraded-goods hedging, but the authors also note that the tradability variable may proxy for information asymmetry. Firms with exports may be more visible and easier for distant investors to evaluate.

6.4 Relation to size and distress anomalies

Small and highly levered firms have been associated with higher average returns in asset-pricing research. The paper suggests that geography and local information may be connected to these patterns.

- Small firms have higher information costs and lower investor recognition.
- Highly levered firms have more uncertain cash flows and higher bankruptcy/distress risk.
- Local investors may be more willing to hold such firms because they believe they understand them better.
- Outside investors may require compensation for adverse selection or uncertainty.

Important caveat. The paper does not claim that geography alone explains the size or distress premium. It suggests a possible link between local ownership, asymmetric information, and cross-sectional return patterns.

7 Short summary

- **Method.** Measure physical distance between U.S. money managers and U.S. firm headquarters; compare actual portfolio distances with benchmark portfolio distances.

- **Headline result.** U.S. managers invest about 160–184 km closer to home than benchmark portfolios imply, corresponding to 9–11 percent local bias.
- **Robustness.** The result survives equal/value weighting, excluding NYC managers, and removing regional, sector, and small-cap funds.
- **Firm types.** Local bias is strongest for small firms, highly levered firms, and firms without foreign sales.
- **Interpretation.** The evidence supports an information-based explanation: proximity may reduce information costs or improve monitoring.
- **Home-bias implication.** Distance alone may account for roughly one-third of international home bias in U.S. equity portfolios.
- **Asset-pricing implication.** Local ownership may be linked to size, leverage, distress, and asymmetric-information effects in expected returns.

8 Conclusion

8.1 Core conclusion

Coval and Moskowitz show that home bias is not only an international phenomenon. Even within U.S. domestic equity portfolios, professional money managers tilt toward geographically proximate firms. The effect is large, statistically strong, and robust across several benchmark and sample definitions.

8.2 Economic intuition

Distance matters because information, monitoring, and familiarity are not frictionless. Local investors may know more about nearby companies, especially when those companies are small, distressed, or locally focused. In those cases, the informational value of proximity is highest.

8.3 Takeaway

The paper reframes home bias as partly a geography problem rather than purely a country-border problem. Investors do not just prefer domestic stocks over foreign stocks; they prefer nearby stocks over distant stocks. This local preference helps explain portfolio composition, provides evidence for distance-based information frictions, and suggests a link between geography and asset-pricing anomalies.